

Multilayer Switch Configuration

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname MLS1

MLS1(config)#ipv6 unicast-routing

MLS1(config)#ip routing

MLS1(config)#VLAN 10

MLS1(config-vlan)#NAME VLAN10_AB

MLS1(config-vlan)#exit

MLS1(config)#VLAN 20

MLS1(config-vlan)#NAME VLAN20_C

MLS1(config-vlan)#exit

MLS1(config)#VLAN 50

MLS1(config-vlan)#name DHCP_Server

MLS1(config-vlan)#exit

MLS1(config)#interface VLAN 10

MLS1(config-if)#

%LINK-5-CHANGED: Interface Vlan10, changed state to up

ip address 192.168.10.1 255.255.255.0

MLS1(config-if)#ipv6 address 2001:DB8:10::1/64

MLS1(config-if)#ipv6 enable

MLS1(config-if)#no shutdown

MLS1(config-if)#exit

MLS1(config)#interface VLAN 20

MLS1(config-if)#

%LINK-5-CHANGED: Interface Vlan20, changed state to up

ip address 192.168.20.1 255.255.255.0

MLS1(config-if)#ipv6 address 2001:DB8:FF::1/64

MLS1(config-if)#ipv6 enable

MLS1(config-if)#no shutdown

MLS1(config-if)#exit

MLS1(config)#interface VLAN 50

MLS1(config-if)#

%LINK-5-CHANGED: Interface Vlan50, changed state to up

ip address 192.168.50.1 255.255.255.0

MLS1(config-if)#ipv6 address 2001:DB8:FF::1/64

MLS1(config-if)#ipv6 enable

MLS1(config-if)#no shutdown

MLS1(config-if)#exit

!Assign ports to VLANs

```
interface range fa0/2-10
MLS1(config-if-range)#interface range fa0/2-10
MLS1(config-if-range)#switchport mode access
MLS1(config-if-range)#switchport access VLAN 10
MLS1(config-if-range)#exit
```

```
MLS1(config)#interface range fa0/11-15
MLS1(config-if-range)#switchport mode access
MLS1(config-if-range)#switchport access VLAN 20
MLS1(config-if-range)#exit
```

```
MLS1(config)#interface range fa0/1
MLS1(config-if-range)#switchport mode access
MLS1(config-if-range)#switchport access VLAN 50
MLS1(config-if-range)#exit
```

! Configure trunk to Router2

```
MLS1(config)#interface fa0/24
MLS1(config-if)#no switchport
MLS1(config-if)#ip address 10.0.0.1 255.255.255.252
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

! Configure OSPFv2

```
MLS1(config)#router OSPF 1
MLS1(config-router)#router-id 1.1.1.1
MLS1(config-router)#network 192.168.10.0 0.0.0.255 area 0
MLS1(config-router)#network 192.168.20.0 0.0.0.255 area 0
MLS1(config-router)#network 192.168.50.0 0.0.0.255 area 0
MLS1(config-router)#network 10.0.0.0 0.0.0.3 area 0
MLS1(config-router)#exit
```

! Configure DHCP pools

```
MLS1(config)#ip DHCP pool VLAN10
MLS1(dhcp-config)#network 192.168.10.0 255.255.255.0
MLS1(dhcp-config)#default-router 192.168.10.1
MLS1(dhcp-config)#dns-server 192.168.60.10
MLS1(dhcp-config)#exit
```

```
MLS1(config)#ip DHCP pool VLAN20
MLS1(dhcp-config)#network 192.168.20.0 255.255.255.0
MLS1(dhcp-config)#default-router 192.168.20.1
MLS1(dhcp-config)#dns-server 192.168.60.10
MLS1(dhcp-config)#exit
MLS1(config)#ip dhcp excluded-address 192.168.10.1
MLS1(config)#ip dhcp excluded-address 192.168.20.1
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
MLS1#write
Building configuration...
[OK]
```

Router Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ipv6 unicast-routing
```

! Interface to Multilayer Switch (already configured)

```
R1(config)#interface Gig0/0
R1(config-if)#ip address 10.0.0.2 255.255.255.0
R1(config-if)#ipv6 address 2001:DB8:FF::2/64
R1(config-if)#ipv6 enable
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3/0, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3/0, changed state to up

R1(config-if)#exit
```

! Interface to Switch3 (PC D)

```
R1(config-if)#interface GigabitEthernet0/1
R1(config-if)#ip address 192.168.30.1 255.255.255.0
R1(config-if)#ipv6 address 2001:DB8:30::1/64
R1(config-if)#no shutdown
```

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
R1(config-if)#exit
```

! Interface to Switch4 (PC E)

```
R1(config)#interface GigabitEthernet0/2
R1(config-if)#ip address 192.168.40.1 255.255.255.0
R1(config-if)#ipv6 address 2001:DB8:40::1/64
R1(config-if)#no shutdown
```

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

```
R1(config-if)#exit
```

! Configure OSPFv2 for IPv4

```
R1(config)#ip routing
R1(config)#router ospf 1
R1(config-router)#router-id 2.2.2.2
R1(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1(config-router)#network 192.168.60.0 0.0.0.255 area 0
R1(config-router)#
00:12:27: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
```

```
R1(config-router)#network 192.168.30.0 0.0.0.255 area 0
R1(config-router)#network 192.168.40.0 0.0.0.255 area 0
R1(config-router)#exit
```

! IPv6 Static Routes to Router1's networks

```
R1(config)#ipv6 route 2001:DB8:10::/64 2001:DB8:FF::1
R1(config)#ipv6 route 2001:DB8:20::/64 2001:DB8:FF::1
R1(config)#ipv6 route 2001:DB8:50::/64 2001:DB8:FF::1
```

! DHCP Configuration

```
R1(config)#ip dhcp pool VLAN30
R1(dhcp-config)#network 192.168.30.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.30.1
R1(dhcp-config)#dns-server 192.168.60.10
R1(dhcp-config)#exit

R1(config)#ip dhcp pool VLAN40
R1(dhcp-config)#network 192.168.40.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.40.1
R1(dhcp-config)#dns-server 192.168.60.10
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 192.168.30.1
R1(config)#ip dhcp excluded-address 192.168.40.1
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write
Building configuration...
[OK]
R1#
```

switch1, Switch2, Switch3, Switch4 (all same basic config) Configuration

!switch 1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Hostname S1
S1(config)#interface range fa0/1-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#exit
S1(config)#end
S1#write
Building configuration...
[OK]
```

!switch 2

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname s2
s2(config)#interface range fa0/1-24
s2(config-if-range)#switchport mode access
s2(config-if-range)#end
s2#write
Building configuration...
[OK]
```

!switch 3

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#interface range fa0/24
S3(config-if-range)#switchport mode access
S3(config-if-range)#end
S3#write
Building configuration...
[OK]
```

#switch 4

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname S4

S4(config)#interface range fa0/24

S4(config-if-range)#switchport mode access

S4(config-if-range)#exit

S4(config)#end

S4#write

Building configuration...

[OK]